Sahan M. Godahewa

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PROFESSIONAL SUMMARY

Materials scientist with 5+ years of experience in computational modeling, molecular simulations, and chemistry research. Skilled in both advanced computation and laboratory techniques, with a strong passion for teaching, mentorship, and translating research into practical applications.

CORE COMPETENCIES

- Molecular modeling and simulation in materials and interfacial systems
- Development, verification, and validation of computational analysis tools
- High-performance computing (parallel processing, cluster environments)
- Molecular dynamics simulations and force field parameterization
- Scientific programming: Python, Fortran, shell scripting
- Advanced statistical data analysis and large-scale data assimilation
- Cross-institutional collaboration with national labs and academic partners
- Technical reporting, peer-reviewed publications, and conference presentations

EDUCATION

University of Kansas - Lawrence, KS
Doctor of Philosophy, Chemistry - ***July 2025***

University of Colombo - Colombo, Sri Lanka
Bachelor of Science (Hons) - ***February 2018***

PROFESSIONAL EXPERIENCE

***Graduate Research/Teaching Assistant*, University of Kansas, Lawrence**
Aug. 2019 – July 2025

- Developed and **validated computational workflows** and tools to functionalize disordered amorphous silica surfaces using graph neural networks with topological ring defects, which reduced the cost and increased functionalization robustness to match experimental silanol densities.
- Served as a **lead Computational Chemist and collaborated** with computational chemists and experimentalists from the University of Kansas and other partnering institutions, and Sandia National Laboratory in understanding 1) the heterogeneity of amorphous silica as a catalyst support and 2) confined fluid dynamics in sedimentary rock formations, leading to multiple publications.
- Applied **advanced statistical concepts,** including t-tests, correlation analysis, non-linear curve fitting, and histograms, to consistently interpret results derived from **computer simulations**.
- **Presented and communicated** research findings at national conferences, delivering novel insights contributing to global scientific understanding.
- **Mentored** and supported a junior graduate student engaged in a collaborative research project in understanding molecular dynamics and quantum mechanical concepts, and also **provided training** on simulation software.
- **Supported undergraduate education** by delivering General (Chem 130) and Physical Chemistry (Chem 537) instructions and facilitating practical lab sessions, and provided feedback to students every week for the lab exercises.
- **Streamlined laboratory operations,** ensuring optimal equipment maintenance, safety compliance, and workflow, cultivating a conducive research environment.

***Advanced Level Mathematics Teacher*, Royal Institute International School, Nugegoda, Sri Lanka**
Feb. 2018 – June 2019

- Taught advanced-level mathematics, including both pure and applied mathematics, to students in Grades 12 and 13, following the local syllabus.
- Designed and implemented mock examinations to enhance student preparedness.
- Achieved a 100% pass rate for students taking the advanced-level exam.
- Attended in-service meetings each term and offered suggestions for areas of improvement.

***Research Assistant*, University of Colombo, Sri Lanka**
Feb. 2017 – Feb. 2018

- Standardized a silica nanoparticle **synthesis** procedure with microwave assistance that decreased the timescale of synthesis, also enhancing the product yield, and **optimized** synthesis conditions resulting in nanoparticles with uniform morphology.
- Conducted amino acid adsorption studies with careful pH balancing and batch processing to identify the adsorption mechanism onto the silica surface, gaining experience on surface characterization methods like **FTIR, SEM, and TGA**.
- **Mentored** a junior researcher in nanoparticle synthesis and extended adsorption studies to the progression of multiple research projects and **provided critical input in team meetings**, shaping the direction of research and subsequent studies that led to a publication and presentation in a research symposium in Sri Lanka.

RESEARCH PROJECTS

Development of forcefield parameters to accurately model silica-CO_2 interactions

- Spearheaded the development of novel van der Waals parameters to be used in classical molecular dynamics simulations for silica-CO_2 systems using benchmark plane-wave DFT calculations.
- Explored novel information about further slowing of the dynamics of CO_2 molecules closer to the surface.

Studying the effect of pore size and pressure on the dynamics of confined CO_2 molecules relevant to applications of sequestration in subsurface environments

- Demonstrated the intensive effects of pore size and pressure and identified the behavior of CO_2 under these conditions using molecular dynamics.
- Quantified the energetic contributions towards the structuring of CO_2 within nanopores using a novel method associated with density profiles.

Ring hydrolysis of amorphous silica to identify the stability of active sites for designing silica-supported catalysts.

- Developed a Fortran-based program to functionalize amorphous silica surfaces using 2-, 3-, and 4-membered rings.
- Compared the hydrolysis energies using ReaxFF to identify the accuracy of forcefield methods in calculating hydrolysis energies.
- Applied Machine Learning Models to predict the ring hydrolysis energy.

PUBLICATIONS

- **Godahewa, S.M.,** Jayawardena, T., Thompson, W.H., Greathouse, J.A. An accurate force field for carbon dioxide-silica interactions based on density functional theory. Journal of Physical Chemistry B, 2025.
- Jayawardena, T., **Godahewa, S.M.**, Thompson, W.H., Greathouse, J.A. Beyond Force Field Mixing Rules to Model Silica-Water Interfaces. Journal of Physical Chemistry C, 2025.
- Senanayake, H.S., Wimalasiri, P.N., **Godahewa, S.M.**, Thompson, W.H., Greathouse, J.A. Ab Initio-Derived Force Field for Amorphous Silica Interfaces for Use in Molecular Dynamics Simulations. *The Journal of Physical Chemistry C* 2023.
- Khan, S.A., **Godahewa, S.M.**, Wimalasiri, P.N., Thompson, W.H., Scott, S.L., Peters, B. Modeling the Structural Heterogeneity of Vicinal Silanols and Its Effects on TiCl4 Grafting onto Amorphous Silica. Chemistry of Materials, 2022.
- **Godahewa, S.M.**, Tillekaratne, A. An insightful approach to understanding the mechanism of amino acid adsorption on inorganic surfaces: Glycine on silica. Chemistry & Chemical Technology, 2023.
- **Godahewa, S.M.,** Jayawardena, T., Thompson, W.H., Greathouse, J.A. Effects of Confinement and Pressure on the Structure and Dynamics of Carbon Dioxide in Silica Slit Pores. Journal of Chemical Physics, 2026.
- **Godahewa, S.M.,** Thompson, W.H. Investigation of the ring size dependence of amorphous silica hydrolysis. Chemistry of Materials (accepted)

PRESENTATIONS

- Development of ab initio-derived force fields for accurate modeling of silica-CO_2 interactions, ACS Spring meeting, 2024.
- Applications of machine learning to predict the surface functionalization of amorphous silica, ACS Fall meeting, 2023.
- Towards accurate modeling of solute-silica interactions using ab initio-assisted forcefields, Kansas Physical Chemistry Symposium, 2023.
- Adsorption of glycine onto silica nanoparticles synthesized by a microwave-assisted method, 6th International Conference on Nanoscience and Nanotechnology, 2019

HONOURS AND AWARDS

- Graduate teaching assistantship, The University of Kansas, KS
- Graduate student travel awards, The University of Kansas, (2023,2024)

SKILLS

- **Programming:** Python, Fortran, Shell scripts, HPC scripting
- **Simulation software packages**: LAMMPS, VASP, NWchem, QChem, CP2K
- **Utility programs**: VMD, PyMOL, MS Office, VESTA, GRACE, LaTeX, Blender, AVOGADRO
- **Experimental techniques:** SEM, TGA, FTIR, UV/VIS Spectrophotometer, pH analysis, Calorimetry
- **Languages:** English(fluent), Sinhala(native)

EXTRACURRICULAR EXPERIENCES

***Vice President*, Sri Lankan Students Association at University of Kansas** (*2021 – 2022)*

- Directed key projects, including the Sri Lankan New Year festival, a summer volleyball tournament, and various fundraising events.

***Alumni Committee Representative*, Department of Chemistry at the University of Kansas** (*2021 – 2022)*

- Organized alumni events, facilitated communication with the alumni board, and presented feedback on behalf of graduate students. Volunteered in planning and executing annual departmental projects.

***Sports Colors Recipient*, University swimming team, University of Colombo, Sri Lanka** (*2014 – 2018)*

- Represented the university swimming team and achieved recognition for excellence in sports.

***Seargent-at-arms*, Rotaract club of Faculty of Science, University of Colombo, Sri Lanka** (*2015)*

- Organized several charity projects that helped children from the underprivileged community.

AFFILIATIONS

- **American Chemical Society (ACS)**
  - Associate member: March 2023-January 2026
- **University of Kansas Chemistry Graduate Students' Organization (CHEM-GSO)**
  - Member: August 2019-July 2025
- **Colombo University Faculty of Science Alumni Association – North America (CUFSAA-NA)**
  - Associate member: August 2019-present